

PROJECT 2: EXPLORATORY DATA ANALYSIS (EDA)

1. Executive Summary & Problem Statement

1.1 Objective

The purpose of this analysis is to conduct a thorough **Exploratory Data Analysis (EDA)** on the transactional dataset comprising **1,200 orders** recorded between **January 1, 2023, and June 30, 2025**. This diagnostic investigation aims to verify data integrity, profile customer purchasing habits, uncover sales performance metrics, and translate mathematical evidence into high-impact strategic business recommendations.

1.2 Key Business Questions

- **Data Integrity:** Is our business data structurally sound and clean?
- **The Center of Gravity:** What does a typical customer spend look like, and how do product choices influence overall pricing?
- **Strongest Predictors:** Which products, payment methods, and marketing channels generate the highest financial margins?

2. Methodology & Data Forensics

To ensure absolute analytical clarity, a structured data cleaning and forensic process was executed on the dataset before running descriptive stats:

- **Completeness Assessment:** Across 1,200 records, the dataset shows **100% data density** across critical tracking fields (OrderID, Date, CustomerID, Product, Quantity, UnitPrice, OrderStatus, ReferralSource, TotalPrice).
- **Missing Values Analysis:** Only the CouponCode variable contains missing values, totaling **309 records (25.8% of the dataset)**.
 - *Forensic Verdict:* This is classified as **Missing Completely at Random (MCAR)**. No treatment or removal is required as these represent organic transactions where a consumer checked out without applying a discount voucher.
- **Data Integrity:** Unit prices, quantities, and item configurations were cross-checked using the formula $\$TotalPrice = Quantity \times UnitPrice\$$. No mathematical corruption was identified.

3. Univariate Analysis: Describing the Distribution

Calculating the "Center of Gravity" reveals how the business actually functions. Below is the summary statistics table for numerical features:

3.1 Five-Number Summary Table

Metric	Quantity	Unit Price	Items In Cart	Total Price
Count	1,200	1,200	1,200	1,200
Mean (Average)	2.95	\$356.41	5.49	\$1,053.97
Minimum (Floor)	1.00	\$11.39	1.00	\$11.39
25th Percentile (Q1)	2.00	\$186.06	4.00	\$410.52
50th Percentile (Median)	3.00	\$364.21	5.00	\$823.62
75th Percentile (Q3)	4.00	\$521.57	7.00	\$1,578.48
Maximum (Ceiling)	5.00	\$699.93	10.00	\$3,456.40

3.2 Key Distribution Insights

- **The Shape of the Evidence:** The TotalPrice distribution is heavily **right-skewed**. While the arithmetic average transaction value is **\$1,053.97**, the median (middle point) is **\$823.62**. This gap proves that average transactional metrics are artificially inflated by high-value, multi-item checkouts.
- **Outlier Diagnosis:** Using the **Interquartile Range (IQR) Method** (defined as $IQR = Q3 - Q1$):
 - $IQR = \$1,578.48 - \$410.52 = \$1,167.96$

- $\text{\$Upper\ Bound} = Q3 + (1.5 \times \text{IQR}) = \text{\$3,330.41}$
- **Identified Outliers:** Only **8 transactions** fell above the \$3,330.41 threshold.
- *Forensic Verdict:* These high-value sales represent legitimate premium checkouts (High Quantity $\times$ High Unit Price) rather than system entry bugs or noise. They should be retained in business reporting to preserve true VIP tracking.

4. Key Findings: Customer & Revenue Signals

4.1 Product Class Performance Summary

By grouping our sales numbers, we see that **Chairs** and **Printers** represent our largest retail drivers:

Product	Total Revenue	Total Units Sold	Average Unit Price
Chair	\$195,620.11	562	\$355.66
Printer	\$195,612.61	542	\$351.71
Laptop	\$192,126.56	535	\$357.71
Tablet	\$186,568.95	497	\$367.68
Monitor	\$175,651.41	480	\$358.66
Desk	\$167,459.93	508	\$329.61
Phone	\$151,722.39	411	\$375.22

4.2 Marketing Channel Attribution (Referral Source)

Social media and digital channels are performing consistently, led by visually engaging platforms:

1. **Instagram:** \$275,285.45 (Top Revenue Source)
2. **Email:** \$261,808.55
3. **Google Search:** \$250,441.48
4. **Facebook:** \$250,410.90
5. **Direct Referral:** \$226,815.58

4.3 Order Status Breakdown

An alarming trend is visible within current operations, as actual completed transactions represent only a small slice of order actions:

- **Cancelled:** 250 orders (20.8%)
- **Returned:** 247 orders (20.6%)
- **Pending:** 237 orders (19.8%)
- **Shipped:** 235 orders (19.6%)
- **Delivered:** 231 orders (19.3%)

5. Strategic Business Recommendations

Based on the forensic evidence gathered during this EDA phase, three high-priority actions are recommended to leadership:

5.1 Address the Order Leakage (Immediate Action Required)

- **The Issue:** Over **41.4% of all logged transactions** end up either Cancelled (20.8%) or Returned (20.6%). Currently, only 19.3% of orders reach a final "Delivered" state.
- **The Recommendation:** A formal operations audit is needed. The high cancellation rate points to a friction-filled checkout process or inventory delays, while the high return rate suggests product quality issues or descriptions that do not match reality. Reducing these leaks by just 10% will save massive amounts in lost margins.

5.2 Build Marketing Campaigns Around High-Margin Products

- **The Issue:** Chairs, Printers, and Laptops are generating the bulk of top-line revenue, while Phones are our lowest-selling category.
- **The Recommendation:** Instead of generic advertising, build targeted marketing campaigns. Pair high-revenue products with our top-performing referral channels—specifically **Instagram** and **Email** marketing—to maximize ROI.

5.3 Optimize Cart Sizes for High-Value Sales

- **The Issue:** Total transaction value shows a strong relationship to unit price ($\$r = 0.72\$$) rather than just raw volume.
- **The Recommendation:** Introduce bundle offers or dynamic product recommendations at checkout (e.g., "*Add a Monitor Stand to your Monitor checkout*"). This pushes average transaction size upward towards our premium customer segments.